- *Applications*—Some applications like eBay, Twitter, transport sites (trains and planes), and pricing/quotes engines emit events as streams of data, usually in real time. We can use Elastic Stack components like Logstash or build in-house, real-time streaming libraries that publish data to Elasticsearch; this data is then used for search functionality.

Elasticsearch expects data to come from persistent storage or in real time to be analyzed and processed internally. The analysis process in the form of search and analytics helps retrieve data efficiently. In a typical setup, organizations use ETL tools to transport data into Elasticsearch. The ingested data is searchable once it makes it through the Elasticsearch store.

3.1.2 Processing data

The basic unit of information in Elasticsearch is represented by a JSON document. For example, we can create a news article as a JSON document for a magazine, as the following listing shows.

Listing 3.1 Typical news article represented as a JSON document

```
{
  "title":"Is Remote Working the New Norm?",
  "author":"John Doe",
  "synopsis":"Covid changed lives. It changed the way we work..",
  "publish_date":"2021-01-01",
  "number_of_words":3500
}
```

COLLECTING DATA

Just as we store data in a relational database, we need to persist (store) data in Elasticsearch. Elasticsearch stores all its data on disk using optimized and compressed techniques. In addition to our business data, Elasticsearch has its own data related to cluster state, transnational information, and so on that needs storage.

If you installed Elasticsearch using a binary (zipped version), it has a folder called data under the installation folder. Data folders can be mounted as network file systems, defined as mount paths, or declared by the `path.data` variable. Regardless of the installation mechanism, Elasticsearch stores data on the optimized file system.

Going a bit further into data folders, Elasticsearch categorizes data per node and per data type. Each node has a dedicated folder with a set of buckets containing related data. Elasticsearch creates a set of buckets (called *indexes* in Elasticsearch lingo) based on each type of data. Simply put, an index is a logical collection of documents. It's a bucket to hold our business data.

For example, news articles may be housed in a bucket called `news` and trades in an index called `trades`, and movies may be split across genre-based indexes such as `classic_movies`, `comedy_movies`, `horror_movies`, and so on. All documents of a particular type (say, cars, trades, or students) end up in their own index. We can create as many indexes as we wish, based on our requirements (keeping in mind memory/size limitations).